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Mu-Ruei Tseng

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Education

Doctor of Philosophy in Computer Science, *Texas A&M University, College Station* — May 2029 (exp.)
Master of Science in Computer Science, *Texas A&M University, College Station* — August 2025
Bachelor of Science in Computer Science & Mathematics, *Hong Kong University of Science and Technology, Hong Kong* — May 2021

Experience

Research Assistant — **Aug 2025 — Present**
PSI Labs, TAMU — *College Station, TX*

- Developed **VOSSA**, a task-aligned speaker representation framework for streaming voice conversion that improves speaker similarity and acoustic realization by learning speaker identity jointly with synthesis, while eliminating a separate speaker encoder, reducing model size by **19%**; first-author Interspeech 2026 paper **shortlisted for Best Student Paper**.
- Developed **AccentCL**, using frozen Whisper-Large-v3 and continual class-incremental adaptation for robust accent classification; **first-author IEEE SLT 2026** paper.
- Co-developed **TVTSyn** (ICLR 2026), **DECRA** (Interspeech 2026), **PHONOS** (IEEE SLT 2026), and **RAFT-Edit** (IEEE SLT 2026) for streaming voice conversion, accent neutralization, and controllable speaker-trait editing.

Research Assistant — **Feb 2024 — Jul 2025**
PSI Labs, TAMU — *College Station, TX*

- First-authored **ECGluFormer**, combining an imbalance-aware multi-objective loss with time-aware Transformer aggregation for glucose prediction from intermittent ECG; outperformed five baselines with 36% missing beats.
- Developed multimodal models for glucose prediction and hypoglycemia detection from ECG, PPG, EDA, and CGM; published first-author work in the *Nature Scientific Data*.

Software Engineer — **Jun 2022 — Jul 2023**
Neurobit Technologies — *Taipei, Taiwan*

- Built a real-time gaze-estimation model running at **30+ FPS** with mean angular error below **5**°.
- Integrated and deployed ML inference and signal-processing pipelines in C#/.NET for gaze, fixation, OKN, vHIT, nystagmus assessment, and automated clinical reporting.

Research Assistant — **Jul 2021 — Dec 2021**
CSE Department, HKUST — *Hong Kong*

- Built a skeleton-based few-shot human action recognition network using geometrically aligned 3D+T skeleton representations.
- Created a 3D+T skeleton generation pipeline using AlphaPose, manual curation, and 3D pose lifting.
- Introduced HAA4D, a clean and class-balanced 4D skeleton dataset for few-shot human action recognition.

Selected Publications (Google Scholar)

First-Author Publications

- AccentCL: Robust Accent Classification with Incremental Expansion — SLT 2026
- VOSSA: Voiceprint Optimization for Streaming Speech Architectures — Interspeech 2026 *(Best Student Paper)*
- ECGluFormer: Glucose Prediction From ECG via Multi-Loss, Transformer-Based Aggregation — IEEE-EMBS BHI 2025
- Hypoglycemia Prediction in Type 1 Diabetes With Electrocardiography Beat Ensembles — JDST 2025
- HAA4D: Few-Shot Human Atomic Action Recognition via 3D Spatio-Temporal Skeletal Alignment — arXiv 2022

Selected Co-Authored Publications

- TVTSyn: Content-Synchronous Time-Varying Timbre for Streaming Voice Conversion and Anonymization — ICLR 2026
- PHONOS: PHOnetic Neutralization for Online Streaming Applications — SLT 2026
- RAFT-Edit: Relative-Attribute Rectified Flow for Static Speaker Trait Editing — SLT 2026
- DECRA: Dynamic Emotion Control for Real-time Speech Anonymization — Interspeech 2026
- A Multimodal Physiological Dataset for Non-Invasive Blood Glucose Estimation — Scientific Data 2025

Technical Skills

ML/AI	PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, LLMs, multimodal learning
Speech/Audio	TTS, ASR, Voice Conversion, Speech Anonymization, Librosa, FFmpeg
Programming/Systems	Python, C++, Java, JavaScript, Docker, AWS, FastAPI, Flask, REST APIs